

MONET Python Function Documentation

Auto-generated

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1 Introduction

This document provides comprehensive documentation for custom functions used in the MONET (Multi Omic clustering by Non-Exhaustive Types) implementation for the MultiOmicsSurvey project. Functions are organized alphabetically within each source file section.

2 run_MONET_no_offset.py

MONET clustering pipeline without correlation matrix offset adjustment. Loads pre-computed correlation matrices and performs multi-omics patient stratification.

2.1 load_correlation_matrices

```
load_correlation_matrices(directory)
```

Description: Load pre-computed correlation matrices from CSV files for multi-omics integration.

Reads all CSV files from the specified directory, expecting exactly 5 files corresponding to different omics data types (e.g., RNAseq, CNV, Methylation, miRNA, SNPs). Each CSV should be a square correlation matrix with sample IDs as both row and column indices.

Parameters:

Parameter	Type	Description
directory	str	Path to the directory containing the CSV correlation matrix files. Each file represents one omic view.

Returns: dict - A dictionary mapping omic names (derived from filenames without extension) to pandas DataFrames containing the correlation matrices.

Raises:

- **ValueError:** If the directory does not contain exactly 5 CSV files.

Example:

```
>>> data = load_correlation_matrices("MO_Corrs")
>>> print(data.keys())
dict_keys(['RNAseq', 'CNV', 'Methylation', 'miRNA', 'SNPs'])
```

2.2 set_seeds

```
set_seeds(seed)
```

Description: Set random seeds for reproducibility across NumPy and Python's random module.

Ensures deterministic behavior for stochastic operations in MONET clustering by setting identical seeds for both NumPy's and Python's random number generators.

Parameters:

Parameter	Type	Description
seed	int	The random seed value to set for reproducibility.

Returns: None

Example:

```
>>> set_seeds(123)
# All subsequent random operations will be reproducible
```

3 run_MONET_with_offset.py

MONET clustering pipeline with correlation matrix offset adjustment. Identical function signatures to `run_MONET_no_offset.py` but operates on offset-adjusted correlation matrices from the `MO_Corrs_offset/` directory.

3.1 load_correlation_matrices

```
load_correlation_matrices(directory)
```

Description: Load pre-computed correlation matrices from CSV files for multi-omics integration.

Reads all CSV files from the specified directory, expecting exactly 5 files corresponding to different omics data types. This version is used with offset-adjusted correlation matrices.

Parameters:

Parameter	Type	Description
<code>directory</code>	<code>str</code>	Path to the directory containing the CSV correlation matrix files (typically <code>MO_Corrs_offset/</code>).

Returns: `dict` - A dictionary mapping omic names to pandas DataFrames containing the offset-adjusted correlation matrices.

Raises:

- `ValueError`: If the directory does not contain exactly 5 CSV files.
-

3.2 set_seeds

```
set_seeds(seed)
```

Description: Set random seeds for reproducibility across NumPy and Python's random module.

Parameters:

Parameter	Type	Description
<code>seed</code>	int	The random seed value to set for reproducibility.

Returns: None

4 Alphabetical Index

Function	Source File
<code>load_correlation_matrices</code>	<code>run_MONET_no_offset.py</code> , <code>run_MONET_with_offset.py</code>
<code>set_seeds</code>	<code>run_MONET_no_offset.py</code> , <code>run_MONET_with_offset.py</code>

5 Notes

5.1 MONET Algorithm Overview

MONET performs patient stratification by:

1. Loading pre-computed sample-sample correlation matrices for each omic
2. Running iterative module optimization (10,000 iterations)
3. Extracting patient clusters from the final module assignments

5.2 Key Parameters

Parameter	Value	Description
<code>iters</code>	10000	Number of optimization iterations
<code>num_of_seeds</code>	100	Number of random seeds for module initialization
<code>num_of_samples_in_seed</code>	10	Samples per seed module
<code>min_mod_size</code>	10	Minimum module size threshold
<code>max_pats_per_action</code>	10	Maximum patients moved per iteration
<code>percentile_remove_edge</code>	80	Edge removal percentile threshold

5.3 Output Files

- `monet_full_output_no_offset.pkl` - Complete results without offset
- `monet_full_output_with_offset.pkl` - Complete results with offset

Each pickle file contains:

- `glob_var`: MONET state object
- `clustering`: Sample-to-cluster mapping
- `all_modules`: Final module dictionary
- `total_time`: Execution time
- `iteration_times`: Per-iteration timing